MIL-C-52053A(ME)

15 February 1985

SUPERSEDING
MIL-C-52053(TC)
18 December 1958

MILITARY SPECIFICATION

CYLINDER: CARBON DIOXIDE, 2 OZ. CAPACITY

1. SCOPE

1.1 Scope. This specification covers a puncturable-seal carbon dioxide-filled cylinder.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8120

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SPECIFICATIONS

FEDERAL

BB-C-101	- Carbon Dioxide (CO_2): Technical and U.S.P.
QQ-P-416	- Plating, Cadmium (Electrodeposited).
TT-I-1795	- Ink, Marking, Stencil, Opaque (Porus and Non-Porus Surfaces).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

MILITARY

MIL-G-20241	- Gasket Material, Wool Felt Impregnated Adhesive, Pressure-Sensitive.
MIL-I-23145	- Inflation Assemblies, Life Preserver.
MIL-P-23377	- Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.
MIL-S-7742B	- Miscellaneous Screw Threads, Standard, Optimum, Selected Series.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedure and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practices for Commercial Packaging. (DoD Adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The cylinder shall be as shown on figure 1 and specified herein.

3.2 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.2.1 Material deterioration and control. The cylinder shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and/or operating environment to which the item may be exposed.

3.2.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation shall be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.2.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

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3.3 Design and construction.

3.3.1 Dimensions. Cylinder dimensions shall be as shown on figure 1.

3.3.2 Gas charge. Each cylinder shall contain 2.10 ounces, plus or minus 0.10 ounce, of carbon dioxide (CO_2) conforming to BB-C-101, grade B.

3.3.3 Piercing area. Each cylinder shall be provided with not less than 5/32-inch-diameter piercing area, as shown on figure 1. The piercing area shall be either an integral part of the cylinder or incorporated in a sealing cap or disc securely welded to the cylinder.

3.3.4 Weight. Gross weight of cylinder shall not be more than 8-1/2 ounces.

3.3.5 Finish. All exterior surfaces shall have a finish as shown on figure 1. Cylinder interior shall be free from scale or foreign matter and shall not corrode.

3.3.6 Plating. Cylinder plating shall conform to QQ-P-416, type II, class 2.

3.4 Performance.

3.4.1 Leakage. When tested as specified in 4.4.2.3 the filled cylinder shall not show any leakage.

3.4.2 Pressure. When tested as specified in 4.4.2.6 the cylinder shall not burst.

3.4.3 Rupture. When the filled cylinder is tested as specified in 4.4.2.7, there shall be no evidence of splintering, fragmentation, or failure at the weld cap.

3.5 Marking. Marking of the cylinder for identification shall be with ink conforming to TT-I-1795 in letters no less than 1/8 inch high, in conformance with MIL-STD-130. Marking on each cylinder shall be as follows:

CYL: CO_2, 2 OZ.
NSN _____
MANUFACTURER'S IDENTIFICATION _____
MFD (Date) _____
TOTAL WT. (nearest 0.1 oz.) _____
U. S.

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3.6 Government loaned property. Unless otherwise specified (see 6.2), the following property will be loaned by the Government (see 6.3).

Description	Identification	Quantity
Inflation Assembly	MIL-I-23145, type II	1

3.7 Workmanship. Workmanship shall be in accordance with standard commercial practice for this type of item. The finished cylinder shall be clean and free from any defects which could affect appearance or functioning. External surfaces shall be free of sharp edges except where sharp edges are required or where they are not detrimental to safety.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications, and standards, as applicable.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3).
b. Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Lot. For the purpose of inspection, a lot shall consist of 1000 cylinders, or any fraction thereof, manufactured under one order, essentially the same conditions, and submitted for inspection at the same time.

4.3.2 Sampling. The sampling for examination and tests shall be in accordance with plans A and B, as applicable.

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4.3.2.1 Plan A. Twenty-five cylinders shall be subjected to each of the following tests:

a. Examination (see 4.4.1).
b. Gas charge (see 4.4.2.1).
c. Heating (see 4.4.2.2).

4.3.2.2 Plan B. Three cylinders shall be selected at random and subjected the following tests:

a. Leakage (see 4.4.2.3).
b. Piercing force (see 4.4.2.4).
c. Internal surface (see 4.4.2.5).
d. Hydrostatic pressure (see 4.4.2.6).
e. Rupture (see 4.4.2.7).

4.3.3 Rejection and retest. A sample cylinder not conforming to any requirement shall be rejected. Failure of any cylinder subjected to the sampling tests shall cause rejection of the entire lot from which the sample was selected. A lot which has been rejected on the basis of piercing force, internal surface, or pressure test may be resubmitted for acceptance, provided five other cylinders selected at random from the lot represented pass the test which had been failed. If any of the group of five does not successfully pass the test the lot represented shall be rejected and shall not be resubmitted for acceptance. A lot rejected on the basis of the dimensional, heating, or leakage tests may be resubmitted for acceptance, provided each cylinder in the lot is inspected and all nonconforming cylinders are removed.

4.4 Inspection procedure.

4.4.1 Examination. Each sample cylinder shall be examined as specified herein for the following defects:

101. Materials not as specified.
102. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and/or operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatments.
105. Dimensions not as specified.
106. Finish not as specified.
107. Identification marking missing, incomplete, or illegible.
108. Workmanship not as specified.

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4.4.2 Tests.

4.4.2.1 Gas charge. Each cylinder shall be weighed prior to and after charging. Nonconformance to paragraph 3.3.2 and 3.3.4 shall constitute failure of this test.

4.4.2.2 Heating. Cylinders shall be heated to a temperature of 160° F., plus or minus 2° F., and maintained at this temperature for not less than 30 minutes. Not less than 24 hours after the cylinders have been heated, they shall be weighed to determine if the gas charge equals the original charge. After heating, each cylinder shall be examined to insure that there is no deformation of the cylinder or piercing area. Failure of cylinders to contain the original gas charge or deformation of the cylinder or piercing area shall constitute failure of this test.

4.4.2.3 Leakage. Three cylinders shall be immersed to a depth of 1 inch, plus or minus 1/4-inch, in a water solution of 1 percent Aerosol OT, clear, 25 percent, or equal. A 20-inch Hg (inches mercury) vacuum shall be maintained above the surface of the solution to determine leakage. Nonconformance to 3.4.1 shall constitute failure of this test.

4.4.2.4 Piercing force. One cylinder shall be installed in an inflation assembly (valve) conforming to MIL-I-23145, type II (see 3.6). The cylinder and valve shall be held in an inverted position, with a 20-pound load attached to the end of the lever arm of the valve. Lever arm shall be held in the inoperative position. When the lever arm is released, the load shall operate the valve to provide maximum penetration of the piercing area by the valve piercing pin. Failure of the piercing pin to immediately penetrate the piercing area shall constitute failure of this test.

4.4.2.5 Internal surface. The cylinder subjected to the test specified in 4.4.2.4 shall be bisected along the longitudinal axis and the internal surface examined. Nonconformance to 3.3.5 shall constitute failure of this test.

4.4.2.6 Hydrostatic pressure. One cylinder shall be subjected to an internal hydrostatic pressure of not less than 8,000 pounds per square inch at room temperature for not less than 5 seconds. Nonconformance to 3.4.2 shall constitute failure of this test.

4.4.2.7 Rupture. One cylinder shall be ruptured by gradually heating in air. There shall be no splintering or fragmentation when the cylinder is ruptured. Complete separation of the piercing area or of any portion of the cylinder assembly from the cylinder shall be considered as fragmentation. If the piercing area ruptures or partially separates from the cylinder, five additional samples shall be selected at random from the lot represented, and subjected to the rupture test. Nonconformance to 3.4.3 shall constitute failure of this test.

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4.4.2.8 Plating. A test to determine compliance with 3.3.6 need not be performed, provided a certificate is furnished certifying that the cylinders have been plated in conformance to QQ-P-416, type II, class 2.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

109.	Quantities boxed together not as specified for level A.
110.	Boxes not as specified for level A or B.
111.	Partitions or separators not provided as specified for level A.
112.	Preservation and packing not in accordance with the referenced document as specified for commercial.
113	Quantities packed together exceed the gross weight or size limitations as specified for level A or B.
114.	Marking missing, illegible, incorrect, or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Six cylinders shall be placed together in a close-fitting box conforming to PPP-B-636, W6c, style optional. Partitions or separators shall be provided between the cylinders. Box closure shall be in accordance with the appendix to the box specification.

5.1.2 Commercial. Cylinders shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

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5.2.1 Level A. Cylinders, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional, in quantities not to exceed a gross weight of 250 pounds. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. Cylinders, preserved as specified in 5.1, shall be packed together in close-fitting boxes conforming to PPP-B-636, V3c, V11c, V13, or V15c, style optional or boxes conforming to PPP-B-640, class 2, style optional. The gross weight or size shall not exceed the limitations of the box specification. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. Cylinders, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. Packages and shipping containers for level A or B shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with ASTM D 3951.

6. NOTES

6.1 Intended use. Cylinder is intended for use in life preserver, parachutist, type B-7, and in other pneumatic equipment requiring cylinder of the type specified herein.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When Government loaned property is not required (see 3.6).
c. Degree of preservation and degree of packing required (see 5.1 and 5.2).
d. Quantity.

6.3 Government loaned property. The contracting officer should arrange to loan the property specified in 3.6, when applicable (see 6.2).

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

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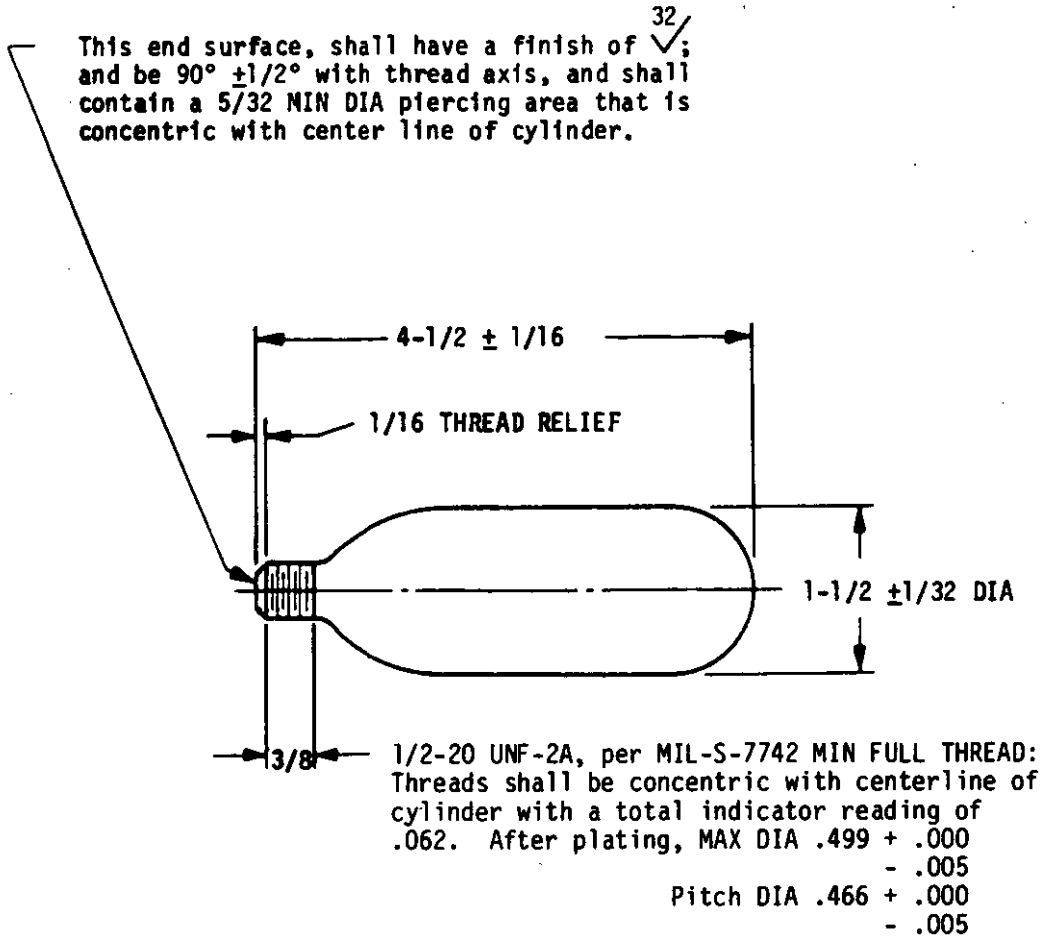
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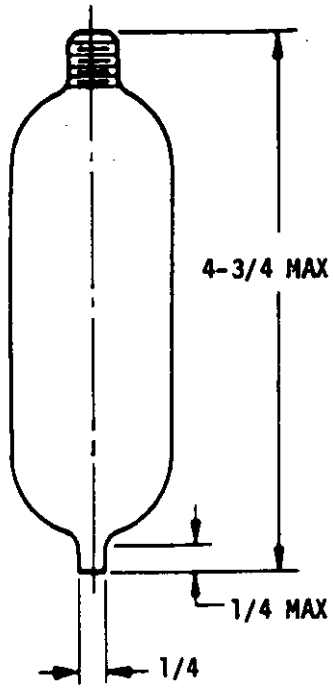
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OPTIONAL DESIGN

Notes:
1. Dimensions are in inches unless otherwise specified.
2. Tolerances on fractions ±1/64, unless otherwise specified.
3. All surfaces shall have 125 finish, unless otherwise specified.

FIGURE 1. Details of cylinder.

X-4074A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-52053A(ME)	Cylinder: Carbon Dioxide, 2 oz. Capacity

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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